

CosmoSort

Next Generation Spatial Cell Sorter

CosmoSort | Spatially-resolved Laser Activated Cell Sorting (SLACS)

An imaging-integrated spatial cell sorter using IR pulse laser-based SLACS technology.



- **Dimensions**
54.5 × 58.3 × 37.8 cm
- **Weight**
Approximately 50 kg

Key Features

Precision at Speed

CosmoSort combines precise sorting with high-throughput performance, enabling target isolation at up to 1 target/second.

Low-Impact Target Isolation

IR pulse laser-based SLACS technology isolates selected targets with minimal sample disturbance for reliable downstream analysis.

From Single Cells to Tissue Regions

Define and isolate targets ranging from 10 μm up to 300 μm , supporting diverse research needs.

Tunable Laser Control

Adjustable laser power and repeat count settings allow users to optimize sorting conditions for various sample types and ROI characteristics.

Broad Compatibility for Spatial Analysis

CosmoSort-isolated samples support diverse molecular applications, from spatial genomics and transcriptomics to proteomics and beyond.

Workflow Overview | Select what you see. Sort what you need



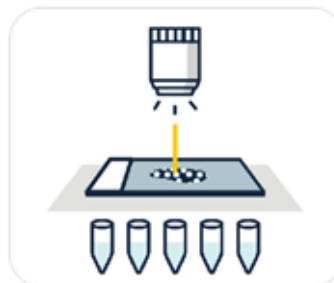
Sample preparation and imaging

Place your sample on a CosmoSlide and apply the appropriate staining method - such as H&E, IF, or IHC, depending on the target of interest. Acquire sample images for target identification.



Target selection

Define ROIs based on morphology, staining patterns, or biomarker expression. Set collection wells and adjust sorting conditions according to the sample and target characteristics.



Spatial cell sorting

CosmoSort isolates selected cells or tissue regions using IR pulse laser-based SLACS technology. Sorted targets are collected for downstream analysis.



Spatial omics analysis

Isolated samples then undergo downstream molecular analysis, linking their molecular profiles to the original spatial context.

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RESEARCH HIGHLIGHTS

From trace samples to spatial multi-omics insight

SPATIAL PROTEOMICS



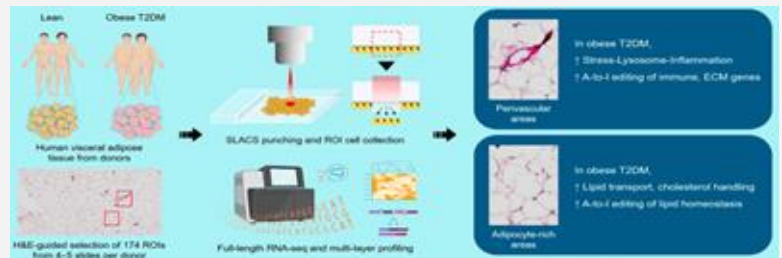
A label-free LC-MS proteomics workflow was optimized for trace cell samples and tissue ROIs. Using CosmoSort-isolated HeLa cells and microscale ROIs from HER2-positive FFPE breast cancer tissue, the workflow identified over 9,500 proteins, revealing spatial proteomic differences between HER2-high and HER2-low tumor regions.

Shin et al. · Analytical Chemistry · 2026

SPATIAL TRANSCRIPTOMICS

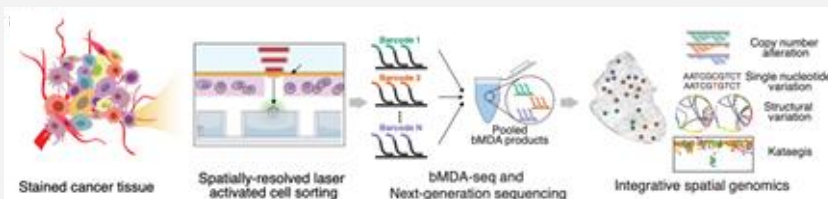
CosmoSort/SLACS-isolated ROIs from archived FFPE human visceral adipose tissue enabled full-length spatial transcriptomic analysis.

The workflow revealed compartment-specific gene expression and obesity/T2DM-associated changes in perivascular and adipocyte-rich regions, including fibro-inflammatory remodeling and altered lipid metabolism.



Jung et al. · Diabetes & Metabolism Journal · 2026

SPATIAL GENOMICS



A barcoded multiple displacement amplification (bMDA) workflow was developed for high-coverage genomic analysis of low-input DNA.

Using CosmoSort-isolated spatial microniches from triple-negative breast cancer tissue, the workflow revealed genomic alterations, kataegis signatures, and spatial subclonal evolution.

Kim et al. · Nature Communications · 2023

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